

Paper Draft

SUBPaper Draft

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Satisfied, Not Solved: What a Demographic Parity Constraint Actually Does

Working draft. Markdown for iteration; ACM LaTeX conversion once the content settles (pdf_lat_ex is available, pandoc is not, so the conversion will be manual).

Every number below is in the repository and re-derived by `tests/test_documented_claims.py`. Related work has been checked against the sources; the evidence trail, including the four references still unverified and why each is low-risk, is in 1. No section is waiting on an experiment.

Scale. 26 populations — Adult, 21 ACS Income populations across twelve states and two protected attributes, and four HMDA mortgage populations across two states and two attributes — over 61 experimental arms once the label-threshold sweeps are counted.

Abstract

Group-fairness constraints of the reduction family (Agarwal et al., 2018) reliably do what they promise: on UCI Adult they drive the demographic parity violation from 0.186 to 0.018 for under two accuracy points, on any base classifier, without modifying the training data. We audit what they do to *get* there, across 26 populations spanning a household survey, its modern replacement, and an administrative record of real mortgage decisions.

(i) Parity is reached by withdrawing favourable decisions rather than extending them: on Adult every method in a six-method ablation shrinks the total by 7.9–22.1%, and 18 of 19 survey populations shrink it. On mortgage data the same constraint *grows* the pool by 4.3%. **The parity metric reports the same success in both cases** — 0.018 where a fifth of all favourable decisions were destroyed, 0.010 where more were created than destroyed. A single-factor sweep, moving only the income cutoff on a fixed population so that rows, features and groups are provably identical across arms, locates the moderator: **the direction tracks the task’s selection rate**, crossing over between 0.25 and 0.60. It replicates across two protected attributes, five states, and fourteen further populations held out from a subsequent prediction ($r = +0.901$).

(ii) A constraint on one attribute leaves intersectional subgroups worse off than any group it was told about, in every sufficiently diverse population and more severely than Adult suggests ($9.0\times$ against $13.2\times$).

(iii) Below roughly 2,500 test subjects the method’s own randomness exceeds the entire effect of the constraint.

(iv) Feature-attribution audits do not recover the mechanism. Three explanations of an apparent +151% attribution shift are each refuted or bounded, and most of the shift proves to be credit

reallocating between two collinear features.

Levelling down is a property of the objective rather than the algorithm, and can be removed by stating the missing objective — a result anticipated in normative form by Mittelstadt et al. (2023), whose minimum rate constraints we position ourselves against explicitly. Our variant is an in-processing moment constraint requiring no protected attribute at prediction time; the exchange rate falls from 1.47 favourable decisions destroyed per one created to 0.88 in all 19 survey populations, and the benefit tracks the damage at $r \approx -0.99$. Against the *optimal* DP classifier the same levelling down appears, so it is not a solver artifact; against minimax group fairness, the established levelling-down remedy, we find it destroys 10% of favourable decisions on mortgage data at an exchange rate of 46.7.

1. Introduction

A fairness-constrained classifier is usually certified by the quantity it was constrained on. The constraint is imposed, the violation falls to some small ε , and the model ships. This paper is about everything that number does not say.

Our starting point is not a criticism of the method. We reproduce Agarwal et al. (2018) on UCI Adult and it does exactly what it claims: demographic parity violation falls from 0.186 to 0.018, on a decision tree and on a logistic regression alike, without a single row of training data being altered. Across six in-processing mitigations the pattern holds. If the question is "did the constraint bind", the answer is yes, everywhere we looked.

The question we ask instead is *how* it was satisfied — and the answer turns out to be invisible to every tool ordinarily used to check.

Contributions.

1. **A four-part audit** of what a demographic parity constraint does beyond its own metric, run identically across 26 populations and 61 experimental arms, spanning two survey instruments and one administrative record of real lending decisions.
2. **A moderator for the most-discussed of those effects.** Levelling down is not what parity constraints do; it is what they do below a selection rate of roughly 0.3, and above it the direction reverses. We establish this by a single-factor sweep holding population, instrument, features, group ratio and proxy structure provably fixed while only the label's cutoff moves, and replicate it across two protected attributes, five states and a second lending domain. **This is our principal claim.**
3. **Evidence that it is not an artifact of the solver.** On Adult the theoretically optimal DP classifier — group-wise thresholding on the unconstrained score — destroys 18.8% of favourable decisions against the reduction's 20.5%, with better parity. At moderate selection rates the two diverge in sign, so the solver matters where the effect is mild and not where it is severe.
4. **An identity that says precisely what levelling down is.** Parity puts both groups on one common rate, so the pie is preserved exactly when that rate's position between the two original group rates equals the privileged group's population share. Levelling down is the optimiser choosing a compromise below the size-weighted one, and nothing else.
5. **An empirical comparison against the existing remedies,** including the finding that minimax group fairness — proposed in the literature as the answer to levelling down — levels down worse than the parity constraint does on data where the parity constraint does not level down at all.

6. **A methodological record.** Predictions registered before results; **three of our own pre-registered predictions failed**; two proposed mechanisms refuted by our own interventions; three claims corrected or retracted in place, including our own novelty claim for the remedy, which the literature had anticipated. Offered as a reason to believe the numbers rather than as process trivia.

What we are not claiming. Neither levelling down nor the remedy for it is a new observation, and we present neither as one — see §2. We also do not claim a mechanism for the moderator: we attempted a derivation, pre-registered it, and report in §4.5 that it does not earn its keep.

2. Related work

Every claim here has been checked against the source, with quoted passages recorded in 0. Four references remain unverified and are listed there with the reason each is low-risk.

Levelling down, and the remedy. Mittelstadt, Wachter and Russell (2023) argue that fairness interventions frequently equalise by degrading the better-off group, and that this is a default rather than an accident of particular methods. **They also propose the remedy:** §6, "Levelling up by design with minimum rate constraints", requires "that every group has, at least, a minimal selection rate, precision, or recall", demonstrates it on Adult with demographic parity, and reports that "levelling down does not occur" while parity is still reached.

We therefore claim neither the observation nor the remedy as novel, and say so here rather than letting a reviewer discover it. Our contribution relative to that paper is four things, each narrow and each checkable:

1. **A scope condition they do not have.** They treat levelling down as a default. We show the direction *reverses* above a selection rate located experimentally. They note in passing that Adult is "more than 75% negatively labelled" — as a remark about accuracy cost, not as a condition — and nowhere connect label prevalence to the direction of the effect. This is our principal claim.
2. **In-processing rather than post-processing.** Their MRC is achieved by post-processing that tunes "a separate offset for each group", which requires the protected attribute at prediction time. Ours is a moment constraint inside the reduction; no model here reads the attribute when predicting. Where per-group thresholds are disparate treatment on their face, that difference is the whole difference.
3. **A population-level floor stacked with parity**, rather than a per-group floor that replaces it — so parity is still enforced to ϵ and the pie is preserved simultaneously.
4. **Held-out replication.** They report Adult, and explicitly on the training set: "Transferring them to the unseen test data introduces noise which would make the results less clear." We report test-set results across 26 populations at five seeds.

Remedies that avoid harm. Minimax group fairness (Martinez et al., 2020; Diana et al., 2021) minimises the worst group's error rather than equalising anything, and decoupled classifiers with preference guarantees (Dwork et al., 2018; Ustun et al., 2019) seek solutions no group would reject. We benchmark against these in §7 rather than only against the unconstrained reduction.

The shape of the optimal fair classifier. Corbett-Davies et al. (2017) show that "for several past definitions of fairness, the optimal algorithms that result require detaining defendants above race-specific risk thresholds"; Menon and Williamson (2018) give the corresponding characterisation as an instance-dependent thresholding of the class-probability function. This gives us a principled

bound rather than merely another baseline, and it predicts that levelling down should not be attributable to the reduction’s search — which §7 tests directly.

The selection-rate axis. The nearest prior work to §4.4 is Goethals, Delaney, Mittelstadt and Russell (2024), *Resource-constrained Fairness* — note that two of its four authors also wrote the levelling-down paper above. It studies the cost of fairness as a function of the available budget across six datasets and rates from 1% to 100%, and reports that "the level of available resources significantly influences this cost, a factor overlooked in previous evaluations". The difference is structural: in their setting positive decisions are a fixed resource, so the total is constant by construction and the directional effect we measure cannot arise. They vary the selection rate and measure how much fairness *costs*; we let the total move and find that its *sign* changes. That the same group studied this axis, named it as overlooked, and did not connect it to direction is the clearest evidence we can offer that the connection is not obvious.

Fairness gerrymandering. Kearns et al. (2018) show that constraints imposed on marginal groups can be satisfied while structured subgroups are badly treated. Our claim (ii) is an empirical replication of exactly that, across populations, with a size condition attached.

Dataset monoculture. Ding et al. (2021) argue the field should stop drawing conclusions from UCI Adult and supply ACS-derived replacements. We take that seriously enough to replicate across 21 ACS populations — and then find it is *not sufficient*, because they share one survey instrument. Our second domain is what breaks claim (i), and no number of additional states could have done it.

Arbitrariness and multiplicity. Black et al. (2022) and work on predictive multiplicity establish that models with equivalent aggregate performance can disagree substantially on individuals. Our claim (iii) is a specific instance, positioned as a caution supported by that literature rather than as a novel finding.

3. Setup and method

Data. UCI Adult (45,222 rows after listwise deletion; Male 30,527 at 31.25% positive, Female 14,695 at 11.36%); ACS Income (Ding et al., 2021) across twelve states, nine of them in both a sex and a race arm; HMDA mortgage applications for Mississippi and Louisiana, 2018, in both arms.

Protocol. Five random seeds throughout, each an independent train/test split stratified on the (*protected attribute*, *label*) interaction rather than the label alone — fairness metrics are read from four cells and the smallest drifts between seeds otherwise, producing variance that reads as model instability but is sampling noise.

The protected attribute is removed from the feature matrix. Fairness through unawareness is carried as a control, not as a method. No model here can read the attribute directly; any gap that survives is one the model rebuilt from proxies, which is what claim (iv) measures.

Metrics are implemented from their definitions and cross-checked against **fairlearn** on every run, because a privileged/unprivileged orientation slip produces plausible and silently wrong numbers. Rates with an empty denominator return undefined, never zero, so an unmeasurable subgroup cannot masquerade as a perfectly fair one — claim (ii) depends on this.

Pre-registration, and what it caught. Where a result could have gone either way, the predictions and their numerical thresholds were written and committed before the experiment ran; the ordering is verifiable in the commit history. It is why we can report that three of our own predictions failed — a mis-specified test of the remedy (§7), a refuted account of the effect’s magnitude (§4.6), and a mechanism that cleared its bars while being beaten by a constant (§4.5). It is also, twice, what caught tests specified so that passing them did not mean what it appeared to mean.

4. The constraint’s effect on outcomes

4.1 Parity is bought by withdrawal

Demographic parity fixes a relationship between two selection rates and is silent on the level at which they meet. On Adult, every one of six in-processing mitigations closed the gap partly by withdrawing favourable decisions, reducing their total by 7.9% to 22.1%. For the reduction under a parity constraint the figure is -20.5% , at an exchange rate of **2.68 favourable decisions destroyed for every one created**.

This is invisible in the certifying metric, which records only that the violation fell to 0.018.

4.2 It replicates across populations, with a diagnostic

Across 19 survey populations, 18 shrink the pool of favourable decisions; the single exception has a degenerate baseline. The rate-level and people-level pictures diverge, and the divergence is predicted by a cross-flow diagnostic at $\mathbf{r} = +0.885$.

4.3 A second domain reverses it

Every population above is a household survey. On HMDA mortgage decisions — an administrative record of real lending outcomes — the same constraint removes 94% of the parity violation while **increasing** approvals by 4.26%, at an exchange rate of 0.50. The sex arm agrees (+1.05%, exchange 0.78), and a second state replicates both. Across-seed standard deviations are 0.17 and 0.29, so the effects clear their own noise by $25\times$ and $3.6\times$.

The parity metric reports 0.010 here and 0.018 on Adult. **Same success, opposite outcome.**

4.4 What sets the direction

The two datasets differ in seven respects at once, so the comparison identifies nothing. ACS Income’s label is "earns more than \$50,000" — a cutoff chosen by the benchmark. Moving it on a fixed state varies the base rate while holding the population, instrument, features, group ratio and proxy structure fixed; we assert in test code that rows, features and groups are unchanged across arms and only the label moves.

cutoff	selection rate	change in favourable decisions	destroyed per created
\$100,000	0.030	-29.71%	22.03
\$70,000	0.099	-22.05%	2.18
\$50,000	0.252	-2.34%	1.14
\$30,000	0.598	$+0.98\%$	0.83
\$20,000	0.760	$+0.80\%$	0.75
\$10,000	0.890	$+0.08\%$	0.89

The direction flips, with the crossover between 0.25 and 0.60. Oregon replicates it more sharply ($r = +0.964$ against Alabama’s $+0.801$) despite starting nearer the crossover; four states of a *second* protected attribute reproduce it at $+0.874$ to $+0.991$; and fourteen further populations, run after and held out from the prediction in §4.5, give $r = +0.901$.

The moderator is the selection-rate *level*, not the between-group base-rate *gap*, and the distinction carries the contribution. That unequal base rates across groups force trade-offs is long established (Kleinberg et al., 2016; Chouldechova, 2017) and is standard textbook material. Our claim is about a different quantity: the overall proportion of the population receiving a favourable decision, irrespective of any gap. The two are confounded in this design by construction — the gap must vanish as the level approaches 0 or 1 — so we partial the gap out, and the relationship *strengthens*, from $r = +0.801$ to $+0.980$ in Alabama and $+0.964$ to $+0.994$ in Oregon.

That control is not a robustness check appended to the result; it is the identification strategy, and it is what separates this claim from the one the field already has.

The sweep places both datasets it was not fitted to: Adult at 0.205 shrinks, HMDA at 0.808 grows. It does not fit them tightly — Alabama loses 2.34% at 0.252 where Adult loses 20.5% at 0.205 — so the selection rate sets the direction but not the magnitude.

4.5 An identity that clarifies, and a mechanism we do not have

Demographic parity puts both groups on one common rate s , so the constrained total *is* s , while the unconstrained total is the size-weighted average of the two group rates. Writing λ for where s falls between them, the pie is preserved **exactly when** $\lambda = p$, with p the privileged group's share. Levelling down is $\lambda < p$ and nothing else — the optimiser choosing a compromise below the size-weighted one. This holds to 0.076 percentage points across fourteen held-out populations and requires no assumptions.

It localises the question without answering it. We attempted a mechanism — under group-wise thresholding, calibration and a location shift, levelling down is a curvature effect and the crossover sits at the mode of the score density — pre-registered it with its thresholds, and tested it on fourteen populations run afterwards. It passes its stated bars, predicting the sign in 12 of 14, and is nonetheless **beaten by the constant rule "levelling up iff the selection rate exceeds 0.5" (13 of 14)**. The measured mode moves inversely with the selection rate ($r = -0.802$), so it carries almost no independent information; controlling for the selection rate, curvature's partial correlation with $\lambda - p$ is -0.182 .

We report the attempt as unsuccessful, and the flip as an empirical regularity. The methodological lesson is recorded with it: we fixed a threshold but not a **baseline to beat**, so clearing it did not mean what it appeared to mean.

4.6 The residual is not group ratio

The obvious candidate for the leftover magnitude was the group ratio, which prior work in this project found to be a genuine cause of the rate-versus-people divergence. We pre-registered the prediction that a larger ratio means more levelling down, and crossed four populations spanning a $4.4\times$ range of ratios with five cutoffs.

It is refuted, with the opposite sign — partial $r = +0.535$ holding the selection rate fixed, against a predicted -0.40 . We do not claim the reverse: the ratio factor has only four distinct levels, enough to refute a predicted direction and not enough to assert its opposite. The magnitude of levelling down remains unexplained, and we report it that way.

5. Subgroups

A constraint imposed on one marginal attribute says nothing about structure inside the groups. Evaluating every Sex $\times$ Race cell, the worst-off subgroup after a sex constraint is treated worse than either marginal group was before it. The effect appears in every sufficiently diverse population and is **worse than Adult suggests** — $9.0\times$ on Adult against $13.2\times$ in Mississippi. It gains one condition: the effect needs a substantial minority to hide in.

In five of ten populations the worst-off subgroup after a sex constraint is a minority man, and in three it is Black men specifically — a finding first observed on Adult and then reproduced without being looked for.

6. What attribution audits can and cannot see

The protected attribute is not in the feature matrix, so the model reaches it through proxies. The natural audit is feature attribution, and the natural expectation — which we recorded in advance — is that a fairness constraint reduces reliance on those proxies.

It does not. On Adult, constraining demographic parity moves SHAP attribution *onto* relationship, whose levels determine sex outright for 46% of rows, by **+151%**.

We then failed, three times, to explain it. The proposed mechanism — that the constraint seeks the best available reconstruction of the protected attribute — was refuted by intervention: a *planted* proxy was used **less** as it sharpened, monotonically. Its replacement, that the constraint is attracted to within-group outcome signal, was refuted by a two-factor version of the same intervention. The third candidate, collinear reallocation, resisted a clean test for a structural reason: holding each column’s marginal informativeness fixed while raising redundancy necessarily lowers the pair’s joint informativeness.

Re-aggregation then showed most of the headline is measurement, not behaviour. Attribution *shares* are compositional; scoring Adult’s two most redundant features as a single coalition takes the effect from **+155% to +11.6%**. The residual is positive in 5 of 5 seeds and is constraint-specific — demographic parity raises the pair’s combined share (+11.7%, +14.2%) while equalized odds lowers it (−26.4%) — which a pure artifact of Shapley credit allocation should not be, since the pair’s redundancy is a property of the dataset.

We report this as a negative result, and the practical lesson is the transferable part: a feature-attribution audit of a fairness-constrained model can move by an order of magnitude without the model’s behaviour changing correspondingly. The +151% is real, reproducible, and unexplained, and attribution did not reveal the mechanism in any of our attempts.

7. Stating the missing objective

Levelling down is a property of the objective, not of the algorithm. The reduction is agnostic about *how* it satisfies a constraint because the constraint is all it is told. **This much is Mittelstadt et al.’s (2023) argument, and their minimum rate constraints are its constructive form;** what follows is a variant and a much larger evaluation, not a discovery.

We add one constraint: $P(h(\mathbf{x}) = 1) \geq \tau$, with τ the unconstrained model’s own selection rate — *do not hand out fewer favourable decisions than the model you are replacing*. Agarwal et al. define constraints as linear in the classifier’s conditional moments, and a floor on the overall selection rate is exactly that, so this needs no new machinery. Unlike a per-group floor applied by post-processing, it requires no protected attribute at prediction time.

On Adult, parity is satisfied to the same tolerance (0.0179 against 0.0178) while the pie loss falls from **−20.5% to −0.6%** and the exchange rate from **2.68 to 1.03**, for 0.37 accuracy points. Across 19 survey populations and both arms, the exchange rate falls in **19 of 19**, from 1.47 to 0.88 and 1.59 to 0.79, and the number of populations creating more favourable decisions than they destroy goes from 1 to 16. The extra accuracy cost averages 0.12–0.15 points.

The remedy scales with the disease. Across the threshold sweep, the amount the floor recovers tracks the amount the plain constraint destroys at $\mathbf{r} = -0.994$ and -0.995 in two states. Where the constraint destroys 29.7% of favourable decisions the floor turns that into +1.2%; where nothing is wrong it does nothing, to within a tenth of a percentage point. That is what makes it applicable by default rather than after diagnosis.

Against the existing remedies. Three populations spanning the selection-rate range, five seeds.

Group-wise thresholding, the optimal DP-constrained classifier, is carried as a **bound** rather than a rival — it uses the protected attribute at prediction time, which nothing else here does. On Adult it destroys **18.8%** of favourable decisions against the reduction’s 20.5%, at a comparable exchange rate and with better parity, which closes the objection that levelling down is a solver artifact. At Alabama’s moderate rate the two disagree in sign, so the solver matters where the effect is mild and not where it is severe.

Minimax group fairness is proposed in the literature as the answer to levelling down. It is not one here. On HMDA it **destroys 10.0% of favourable decisions at an exchange rate of 46.7** — the worst figure anywhere in our data — while barely moving parity, on the same population where the parity constraint *creates* 4.3%. The reason is in its objective: minimax minimises the worst group’s **error**, and worst-off-by-error is not worst-off-by-outcome. On Adult the higher-error group is Male, so the method optimises for the privileged group and concludes correctly that there is little to do. This is our own thesis appearing inside a baseline — a criterion whose stated aim diverges from what it optimises, with nothing in its metric to say so.

What the floor is, after this comparison: not the best at parity, and not the cheapest. It is the only arm that satisfies parity *and* preserves the pool of favourable decisions *without* requiring the protected attribute at prediction time.

8. Limitations

- **Two lending populations, one year.** The reversal in §4.3 rests on HMDA 2018 for Mississippi and Louisiana. A second high-selection-rate *domain*, not merely a second state, is needed before the crossover is quoted as a property of tasks in general.
 - **The magnitude is unexplained.** The selection rate sets the direction, not the size; an order of magnitude of variation at comparable rates remains open after one candidate was refuted (§4.6) and one derivation failed to earn its keep (§4.5).
 - **No direct comparison against a post-processing MRC.** Mittelstadt et al.’s construction is a per-group floor applied by post-processing. The closest arm here is group-wise thresholding, which is not the same thing. This is the most obvious missing baseline.
 - **The mechanism behind §6 is unidentified,** and we say so rather than offering a fourth story. Our record with plausible mechanistic accounts is three proposed, two refuted by intervention and one bounded by re-aggregation.
 - **HMDA’s feature set is a judgement.** Thirty-six columns were excluded as post-decision or protected. `denial_reason` is 99.2% predictive of the outcome with a missingness gap of exactly zero; a different exclusion list would give different numbers.
 - **Minimax is a simplified implementation** — the standard reweighting scheme rather than either paper’s exact algorithm — so its erratic behaviour may be partly implementation. The Adult diagnosis, that the higher-error group is Male, does not depend on it.
 - **Five seeds,** which our own claim (iii) suggests is too few on the smallest populations.
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9. Discussion

Fairness constraints do exactly what they are asked. Every finding above follows from the constraint being satisfied precisely as specified, by a solver doing its job. The gap is between what was asked and what was meant, and none of the standard toolchain surfaces it: not the certifying metric, which reports the same success for opposite outcomes; not marginal parity, which is silent on

subgroups; not a point estimate, which hides run-to-run variation exceeding the effect; and not feature attribution, which moved ninefold without telling us why.

The practical reading is narrow and, we think, actionable. If a deployment cares whether the pool of favourable decisions shrinks, that has to be written into the objective, where it costs a fraction of an accuracy point. If it cares about subgroups, marginal parity will not deliver them. And if the task’s selection rate is low, the constraint’s default behaviour is to take decisions away — which is, in lending, hiring and admissions, exactly the regime these methods are proposed for.

We close on what we could not do. We can say *when* the direction flips and not *why*. Our own derivation of the crossover cleared the bars we set for it and was beaten by a constant, which we report because a fairness literature that publishes only its successful mechanisms is subject to precisely the criticism this paper makes of fairness metrics: the summary looks the same whether or not the thing beneath it worked.

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